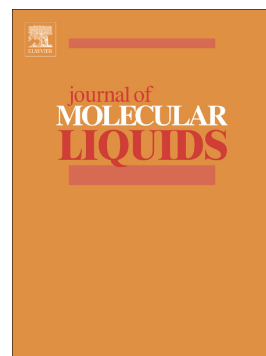


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Nanoconfined Ionic Liquids: A Computational Study

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Abstract

In this work we study how the properties of ionic liquids change upon confinement between two walls at various nanometric distances. We also find a correlation between the dynamic properties and the structure of this system under these extremely nanoconfined circumstances. We report that for ethylammonium nitrate, the ionic structure inside the simulation boxes is a consequence of a subtle combination of ion packing, hydrogen bonding, electrostatic and van der Waals forces. We also study how the dynamical properties of this material change in these conditions, showing a decrease in the ionic mobility for very low slit sizes, recovering the bulk behavior as the distance between walls increases. Moreover, when short wall separations are considered, we show that the ionic mobility at short times is minimum when the hydrogen bonding is also minimum. Finally, we show that the effect of the interface in the long time ion dynamics only extends to a quite small region (ca. 1.35 nm) next to the walls. We show that the overall behavior of the nanoconfined liquid slit can be described by the superposition of a bulk and a confined simulation with a fitted wall separation. To our knowledge, there are no similar studies characterizing this kind of dense ionic systems in such a fine-grain way under these ultra-nanoconfinement conditions.

Keywords: Ionic liquids, nano-confinement, structure, dynamic-structure correlation, molecular dynamics simulation

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1. Introduction

In the last decades, ionic liquids (ILs) have proven to be materials with a wide range of applications: as green designer solvents for any kind of solutes, as media in catalytic reactions, as lubricants, electrolytes in electrochemical devices, etc. [1–10]. For many of these applications, a thorough understanding of IL properties close to the interface is crucial, so huge efforts have been made to this line both experimentally and computationally (see Ref. [11] and references therein). Many of the interesting interfacial properties of these materials are not accessible through experiments and very often computational methods such as density functional theory calculations or molecular dynamics (MD) simulations are the only way to access those properties or to complement experimental measurements. Indeed, many articles studying ILs close to interfaces using these techniques can be found in the literature, e.g. characterizing the interface between ILs and a gas or vacuum [12, 13], in presence of electrochemically relevant surfaces [14, 15] or even the liquid-liquid interface [16].

Very frequently, ILs are used under nanoconfinement, so the understanding of their behaviour close to solid interfaces is crucial (see Ref. 17 for a complete review of this subject). One can find in this line very interesting studies about the application of these dense ionic fluids in presence of different materials such as nano-porous silica-based matrices [18, 19], carbon nanotubes [20–22], polymers [23, 24] or even forming ionogels [25, 26]. Putting ILs in these conditions opens the door to develop technologies that benefit from the enhanced dynamics of ions in the liquid but integrated in rigid devices.

To access the molecular details that allow us to understand the physics in these systems, computational methods are needed. Nowadays, we can find in the literature a variety of works with detailed descriptions of the molecular structure in this kind of environments [27–36]. However, the dynamic properties, which are very important due to the interest in preserving the diffusion and the conductivities of ILs in this confined media, are confusingly described in the literature, specially when it comes to computational studies. For instance, if we focus in the diffusion of ILs in carbon nanotubes, there are contradictory experiments that report both faster [37] and slower [38] diffusion of these liquids comparing with the results in bulk liquids. Moreover, an increase by a factor of 3 relative to the value in the bulk is registered for [OMIm][BF₄] IL immersed in a matrix of nanotubes, while no changes are observed in the same conditions for [BMIm][NTf₂] (a less

self-organized IL) [37]. The disruption of the hydrogen bond network due to the presence of interfaces is believed to be the cause of the enhancement of the diffusion mechanism. On the other hand, there is not either a well-established theory that explains the preservation of the IL conductivity when they are forming ionogels inside nano-porous materials [39, 40]. Although there are some computational works that tried to provide a theoretical background to explain these phenomena [41–44], more research must still be done in this field.

Recently, in this line, Duan *et al.* [45] investigated the properties of liquid aluminum confined between different substrates with different slit sizes using MD simulations. They reported a correlation between axial layering of liquid aluminum, which is determined by the slit size, with the diffusion coefficients in the direction parallel to the interfaces as well as with other structural properties. They found that the diffusion coefficient of aluminum atoms shows maxima for slit sizes just a little bit lower than those where the layering transition begins, i.e. when the number of aluminum layers between the interfaces goes from n to $n+1$. Due to the layering in ILs associated to charge oscillation (a region of inhomogeneous distribution of charge commonly known as electric double layer), this correlation may explain some properties of these systems under confinement.

Nevertheless, for ILs, it is hard to provide accurate predictions regarding the dynamic properties using computational methods. This is mainly due to the usage of non-polarizable force-fields, which are known to underestimate the diffusion of the components of the simulated ILs [46–48]. The packing of the components in the simulation box may also greatly influence the dynamic properties [42]. Even so, although the dynamics simulated using these force-fields will not be quantitatively correct, they are usually considered to be accurate enough to perform qualitative studies [46].

In this work we present MD simulations of one of the most prototypical protic IL, ethylammonium nitrate (EAN), confined between two interfaces with different separations (from 1 nm to 8 nm), in order to study the effects of nanoconfinement in the structural and dynamical properties. Moreover, following the work of Duan *et al.* [45], the precise correlation between IL properties with the distance between walls, lateral slit size and ionic layering between the walls is studied. We also study the spatial range of the perturbation that the wall exerts on the liquid using a mean-field-like theory that allows us to describe a simulation with ILs between two walls as the superposition of a bulk and a highly confined system with a characteristic wall separation. This

characteristic length provides a reference of the scale at which the properties of ILs are expected to change due to the confinement, which may help to the understanding of ILs properties in confinement conditions such as ionogel matrices or in nano-porous electrodes.

In the following section we are going to introduce all the details and parameters that we use in our simulations. In the next sections the obtained results are presented and discussed. Finally, our main conclusions are summarized in the last section of the paper.

2. Simulation Details

We performed MD simulations of EAN IL using the OPLS-AA force-field [49] by means of the Gromacs 2019.3 software [50, 51]. The employed molecular topology for EAN, which was optimized to reproduce the experimental density, has been reported previously [52]. For the interest of the reader, an example of the topology files used in our simulations is shown in the first section of the Supplementary Information (SI). Regarding the interfaces, as we were not specially interested in the wall-IL interactions and to reduce the computational costs, implicit walls were used, i.e. instead of simulating all the atoms that may constitute a real material, the interactions are averaged in an interaction potential of two planes perpendicular to the Z direction. The chosen atom type for these walls was carbon `cpls_145`. To calculate the interaction with the walls, a surface-integrated Lennard-Jones potential was chosen. Although it does not affect the simulation results as these implicit interfaces are forced to be at fixed positions, the atomic wall density was chosen to match the graphene one ($38.18 \text{ atoms/nm}^{-2}$). The simulation time-step was 1 fs and periodic boundary conditions (pbc) were applied only in the X and Y dimensions. All the simulations were performed in the NpT ensemble at the pressure of 1 bar, allowing fluctuations in the volume only in the X and Y directions and thus keeping a constant slit size maintaining the walls in fixed Z positions. This can be achieved imposing a semi-isotropic pressure coupling, which allows us to simulate different confinement scenarios. See section 2 in the SI for a more detailed description of the simulation parameters.

Initially, using Packmol [53], the molecules for the simulations were randomly packed inside a box with the desired Z and estimated X and Y dimensions to preserve the bulk density, e.g. in a first set of simulations, 500 ionic pairs were packed in simulation boxes with slit sizes of {1, 1.1, 1.2, 1.3, 1.4, 1.5, 1.6, 1.7, 1.8, 1.9, 2, 3, 4, 5, 6} nm and 1000 ionic pairs in a box with a wall

separation of 8 nm (to prevent the lateral dimension to be shorter than the Coulomb interaction cutoff length). Then, two energy minimizations, a first one with the steepest decent method and the second one with the conjugate gradient algorithm, were performed to relax the artificially obtained molecular configurations. After that, a simulated annealing of 1 ns from room temperature to 600 K and then back to room temperature was performed to allow the systems to explore wider regions of their energy landscapes and thus, to prevent not properly stabilized configurations. The temperature increased/decreased linearly during the heating/cooling process during 0.5 ns. Once the annealing step had finished, stabilization runs of 10 ns were performed in order to allow the systems to stabilize at room temperature. Finally, production runs of 30 ns were performed to record the position of the simulated atoms at each picosecond. These positions were used to obtain the results shown in this manuscript. Additionally, for some of the systems, short simulations of 50 ps were performed recording the velocity of each atom in all the simulation frames to study the velocity autocorrelation function (VACF).

3. Results and Discussion

In a first attempt to study the correlation between the axial structure with the diffusion of ILs in confined environments, we performed a set of simulations with different slit sizes. With these simulations (those mentioned in the previous section), and excluding the system with 1000 ionic pairs, the volumes of the simulation boxes are seen to remain constant (apart from neglectable fluctuations in the ionic density due to confinement restrictions) for all the systems as the lateral box size changes to fit the pressure fixed by the barostat. In Fig. 1 we plot the number density profiles along the Z axis for anions and cations in these simulations. In this figure we can see how in the most confined simulations (slit sizes from 1 to 3 nm) charge oscillations due to the alternating layers of anions and cations are presented across the whole simulation box. This is not the case for less confined simulations where a central region without charge oscillations (bulk-like region) can be distinguished. This is in agreement with previously reported simulations that formed a spatial extension of the electric double layer of 1-2 nm [27, 30, 54, 55]. Another interesting phenomenon is that the shape of the ionic distribution nearby the interfaces is practically the same independently of the distance between walls. This ionic distribution matches with previously reported results for similar systems [55–57] although the majority of those results

were obtained with simulations in the NVT ensemble. However, our simulations with slit sizes from 1 to 2 nm present a novelty as, to our knowledge, there are no works in the literature that study how the ionic layers appear as the slit size increases in such a fine grained way specially confined between neutral interfaces.

Figure 1: Number density profiles for anions (blue) and cations (red) for the simulations performed changing the distance between interfaces. The slit size of each graph can be read in the Y axis. These profiles can not be quantitatively analyzed as they are a projection of a 3D plot so the height of the curves are scaled differently. However, they are accurate enough to compare the ionic distribution along the Z axis as the slit size changes.

Duan *et al.* [45] reported that the diffusion of liquid aluminum in nanoconfined environments (similar conditions as in our simulations) is highly affected by the layer formation between the interfaces. They also characterized the layer formation studying the potential energy of the systems as well as the volume occupied by the aluminum atoms. With the same purpose we represent in Fig. 2 the densities and the potential energy. It has to be mentioned that the represented number densities are the results of dividing the number of molecules in the systems by their accessible volume in the simulations boxes (two slices of 0.2 nm width are subtracted from the total volume due to the exclusion region associated to the implicit walls Lennard-Jones parameters). In general, in the most confined geometries, there are less molecules per unit volume, i.e. a lower density. As the distance between the walls increases, the bulk density is recovered. This can be explained arguing that the presence of walls avoids an efficient packing of the ions. However, at some well defined slit sizes ($\{1.1, 1.5, 1.9\}$ nm), local maxima of the density are registered. This takes place when the distance between the interfaces allow densely packed ionic configurations minimizing the free space. This same behavior has been already reported for hard disks and spheres packing in confined geometries [58]. Moreover, we can see that the potential energy takes minimum values in the slit sizes for which the densities are maxima, pointing out to highly organized structures. We can say, as it was previously introduced in the literature for ILs confined inside nanotubes [59] or between gold planar interfaces [60], that the IL is crystallized when these wall separations are simulated. Which is more interesting and agrees with the observations of Duan *et al.* [45] is that the slit sizes corresponding to the peaks in these curves are

associated to the formation of new ionic layers between the walls. Indeed, the first peak, which corresponds to a distance between walls of 1.1 nm, shows up when an increase of 1 Å in the distance between walls in the system with 1 nm of separation allows the formation of a second layer of cations (see Fig. 1). The other two peaks placed at slit sizes of 1.5 nm and 1.9 nm match with the transition from two to three and from three to four anion layers, respectively.

Figure 2: Total number density (blue) and potential energy (red) for the simulated systems with 500 ionic pairs changing the distance between walls.

To find out if there is any correlation between the ionic layering and the ionic diffusion we calculated the self-diffusion coefficients (D) using the conventional Einstein relation:

$$D = \lim_{t \rightarrow \infty} \frac{\langle \Delta r^2(t) \rangle}{6t} \quad (1)$$

and calculating the mean squared displacement (MSD, $\langle \Delta r^2(t) \rangle$) from the position of the molecular center of mass. However, we found that, although the same IL is used in all the simulations, not all the systems reach the diffusive regime in the simulated time, showing a quite important dependence with the distance between walls. One of the most used methods to check if the simulation components reach the diffusive regime is to plot in log-log scale the MSDs and determine if the curves reach unit slope at some time. With this aim, we plot the anions' MSDs for the simulations with slit size of 1, 2, 3, 4, 5, 6, 8 nm and compare them with the anions' MSD in bulk conditions in Fig. 3. The unit slope line is also represented in this figure. We can see that for the less confined systems, specially for those with 8 nm of distance between walls, the MSD becomes very close to that in bulk conditions at long times and thus it reaches the diffusive regime. The MSD curves were fitted in the time interval from 10 to 20 ns to the following equation:

$$\langle \Delta r^2(t) \rangle = 2\sigma^2 + 6Dt, \quad (2)$$

where, as it was mentioned, the diffusion coefficient describes the long time behavior of the MSDs and σ is related with the short-time fluctuations in the molecular position (see Ref. 61 for the derivation of this equation).

Figure 3: MSDs in log-log scale for the anions in the simulations with slit sizes of 1, 2, 3, 4, 5, 6, 8

nm and 500 ionic pairs (excluding the 8 nm simulation which was performed with 1000 ionic pairs). The MSD for the anions in bulk conditions is also represented (solid black) as well as the unit slope line (dashed black).

With these considerations, we represent in Fig. 4 the anions' MSDs in linear scale as well as the corresponding descriptors for the set of simulations whose density profiles are represented in Fig. 1. In this figure we can see the evolution of these magnitudes with the distance between walls and how they compare with their values in bulk simulations. The ion mobility increases as the distance between the walls increases, which was also pointed out in Fig. 3. If the values for simulation with slit distances between 1 and 2 nm, which will be analyzed in the following lines, are neglected, we can see that the diffusion coefficients monotonically increase from ca. $1.3 \text{ \AA}^2/\text{ns}$ to its maximum in the simulation with slit size of 8 nm at ca. $7.8 \text{ \AA}^2/\text{ns}$, which is very close to its value in bulk simulations ($\sim 7.3 \text{ \AA}^2/\text{ns}$). So, in general lines, we can say that the confinement induces lower ionic mobilities. This is in agreement with previously reported results for imidazolium-based ILs confined between graphene [36] and graphite [62] walls. This trend is not so obvious for the fluctuations in the positions at short times σ , although we can still assert that it takes lower values in most confined simulations, i.e. the amplitude of the ionic position oscillations are lower. However, paying attention to the results for the most confined simulations, specially for those with slit sizes between 1 and 2 nm, strongly fluctuating results are registered.

Figure 4: (a) Anions' MSDs for all the performed simulations with 500 ionic pairs as well as the one for the system with 1000 ionic pairs and 8 nm of distance between walls. (b) and (c) summarize the dependence of the diffusion coefficients and position fluctuations, respectively, obtained fitting the MSDs between 10 and 20 ns (shadowed region) to the Eq. 2. The results for bulk simulation are also represented in black dashed lines for comparison.

In order to achieve more reliable results for the dependence of the diffusion coefficient with the slit size of the simulations we need to consider some effects derived from the simulations parameters that may induce undesired fluctuations in the results, specifically the dimensions of the simulation boxes. There is an artificial effect that has nothing to do with the used force-field that affects the dynamics of the simulations which is related with hydrodynamic interaction of the

simulation components with their replica produced by the pbc's. This phenomenon has been firstly proved by Yeh and Hummer [63] to be more significant in small simulation boxes for bulk conditions for water and Lennard-Jones fluid simulations. They found that for those systems the self-diffusion coefficients decrease linearly with the inverse of the simulation box length. Then, Gabl *et al.* [64] showed that these theoretical predictions can also be applied for IL simulations but only for large enough boxes. The minimum box length required for the application of this theory depends on the viscosity of the studied fluid. For their systems, the authors conclude that at least 500 ionic pairs are needed to simulate boxes large enough so as the dynamical properties are well behaved. They also reported that the structural properties are much less affected by the simulation box size. More recent works [65, 66] describe how the theoretical correction due to this hydrodynamic effect has to be modified to predict the diffusion coefficients for non cubic simulation boxes. Although these results consider the problem of rectangular simulation boxes, their simulations had pbc's in the three dimensions. Recently, Simonnin *et al.* [67] solved the hydrodynamic problem for confined simulations, i.e. a Lennard-Jones liquid between two rigid walls with pbc's only in directions parallel to the interfaces X and Y. Their theoretical correction to the diffusion coefficients fits very well with their results from the simulations. However, to our knowledge, this study has not been extended to realistic molecular simulations yet.

Figure 5: Structural descriptors for the simulations changing the number of ionic pairs with 1.7 nm of wall separation: (a) density profiles for anions and cations along the Z axis; (b) coordination number of anions around cations as a function of distance; (c) number of hydrogen bonds per molecule. The dotted black line represents the average number of hydrogen bonds per ionic pair in bulk simulations and the shadowed region covers its standard deviation.

Up to now, in the reported simulations, the number of ionic pairs remains constant (500) as well as temperature and pressure. On the other hand, the distance between walls is forced to change as we want to study the effect of the slit size. This means that the lateral dimensions of the simulation boxes must change to compensate the pressure. That is why the lateral box size in the simulation with 1 nm of wall separation is around 11.9 nm and around 3.66 nm for the simulation with a slit size of 6 nm. This variation in the lateral dimensions may be the source of important changes in the ionic diffusivity as it was mentioned. To check the effects that the simulation lateral

box size has on the dynamic properties of the system under study, we performed a new set of simulations changing the number of ionic pairs and keeping constant the distance between walls ($S = 1.7$ nm). Thereby, we used simulation boxes with lateral sizes from 6 nm (for the smallest system with 300 ionic pairs) to 24.6 nm (for the largest one with 5000 ionic pairs). In Fig. 5 we summarize the main structural magnitudes for the simulations: the axial density profiles for anions and cations, the coordination number of anions around cations and the average hydrogen bonds per molecule. Looking at this representation we can conclude that indeed, as Gabl *et al.* [64] reported for bulk IL simulations, the simulation box size does not significantly affect the structure. However, this may not be true for the dynamics. In Fig. 6 the MSDs of the anions as well as the descriptors obtained from the fit to Eq. 2 for these new set of simulations are represented. The MSDs are not even close to fall in the same curve as it was the case for the structural properties. On the other hand, the diffusion coefficients do not show a clear correlation with the simulation box sizes as it was reported for bulk IL simulations [64] or for confined Lennard-Jones fluids [67]. It seems that there are fluctuations of the diffusion around an average value (ca. $0.8 \text{ \AA}^2/\text{ns}$, see Fig. 6-b) which are damped out as the simulation box size increases. In contrast, in bulk simulations the diffusion coefficients for small simulation boxes change randomly (non-linear region), but for larger boxes (more than 500 ionic pairs or box length larger than 5 nm) the coefficients linearly increase with the box side length (linear region). If we compare with the results for simulations of confined Lennard-Jones fluids obtained by Simonnin *et al.* [67], we see that our results are compatible if we compare them with the corresponding systems (small wall separations compared with the lateral size). They also reported that in this highly confined simulations, where it can not be found any bulk-like region, the corrections due to hydrodynamic interactions with the simulation replicas can not be applied. This is related with the fact that the liquid in these geometries can not be considered as a continuum fluid, so Navier-Stokes equations are not properly applicable. Moreover, this is enhanced by the nano-structured nature of ILs. Hence, a new theory for these fluctuations is needed. Our guess is that this new theory should consider collective long-range molecular interactions between molecules and their replicas. However, the development of this formalism is out of the scope of this article and will be properly addressed in future works. Going back to the results, it has to be mentioned that the fluctuations in the diffusion coefficient in our system are not so important if they are compared with the value of the diffusion coefficient for bulk simulation (as it was mentioned, around $7.3 \text{ \AA}^2/\text{ns}$). However, they are

relevant enough if we want to study their behaviour in highly nanoconfined geometries. The same applies to the fluctuation of the ions positions at short times (see Fig. 6-c). Consequently, lateral size must be considered if we want to safely compare the results in confined environments.

Figure 6: (a) Anions' MSDs for the simulations with 1.7 nm of separation between walls and different number of ionic pairs. (b) and (c) represent the dependence with the inverse of the lateral simulation box size of the diffusion coefficients and position fluctuations, respectively, obtained fitting the MSDs between 10 and 20 ns (shadowed region) to the Eq. 2.

Taking into account these results, we studied the correlation between sets of simulations changing the distance between walls but now fitting the amount of ions to keep constant the lateral simulation box size. We chose a simulation box side around 9.5 nm, looking at the results presented in Fig. 6, which is large enough to avoid big fluctuations in the dynamic magnitudes while keeping an acceptable computational cost. The slit size in these simulations goes from 1 nm (321 ionic pairs) to 3 nm (1562 ionic pairs) in intervals of 0.1 nm. We also performed a simulation with wall separation of 8 nm (4600 ionic pairs) to see how the dynamical magnitudes change for large slit sizes. As we did for the simulations with constant number of ions, we started the analysis representing the axial density profiles mainly to compare them with the previously obtained ones and to see the slit sizes corresponding to new ionic layer formation. These densities are represented in Fig. 7. Clearly, the curves for systems with slit sizes from 1 to 2 nm are identical to those in Fig. 1. So, once again, the structural properties do not depend on the simulation box size. Regarding the slit sizes from 2 to 3 nm, it is clearly shown that the new ionic layer formation in the middle of the boxes does not change at all the ionic distribution close to the interfaces. For these new slit sizes we find two new wall separations at which the layers of anions are better defined: 2.3 and 2.7 nm.

Figure 7: Same representation as in Fig. 1, but now for simulations performed varying the distance between interfaces from 1 to 3 nm keeping the lateral simulation box size constant.

Figure 8: (a) Total number density (blue) and potential energy per ionic pair (red); (b) number of hydrogen bonds per ionic pair for simulations with constant lateral box size. The dotted black line represents the average number of hydrogen bonds per ionic pair in bulk simulations and the shadowed region covers its standard deviation.

Following the same procedure as for the first set of simulations, we represent the system's number density and its potential energy per molecule in order to better locate the ionic layer formation. We also calculated the hydrogen bonds per molecule to see how they are modified relative to its bulk value and the results are plotted in Fig. 8. Once again, a clear correlation between the density and potential energy is observed, showing maxima (minima in case of potential energy) in the simulations with the most densely packed configurations, i.e. in systems with a well-defined ionic layering. On the other hand, minima in the density are associated to more disordered configurations (see the ionic density profiles in Fig. 7 for the system with a wall separation of 1.4 nm, which shows a local minimum in the global density). Moreover, apart from the minima already pointed out for the first simulations at slit sizes of 1.1 nm, 1.5 nm and 1.9 nm, we can see a new one in 2.3 nm and another, but much less marked, around 2.7 nm. These distances match with the ones predicted looking at the density profiles, which supports our hypothesis that states that more ordered systems are those which minimize the potential energy, hence minimizing the free volume and maximizing the density. If we take a look at the number of hydrogen bonds (Fig. 8-b), a very interesting correlation with the ionic layering is observed. Indeed, the average number of hydrogen bonds per ionic pair shows minima (whose values are close to the average value in bulk conditions) in more packed systems (maxima in the density which are also values close to the bulk simulation). Moreover, if these minima are not taken into account, the number of hydrogen bonds in the studied IL is larger in confined conditions than in bulk. In other words, in highly confined environments, where the ions can not rearrange to minimize the free volume, the hydrogen bond network is reinforced unless the distance between walls allows the IL to crystallize. This is a very interesting finding as some previously reported results point out the importance of the hydrogen bond network in the dynamical properties of confined ILs (see the discussion in section 7.2.2 in the review by Zhang *et al.* [17]). Our results show that the slit size of the pores in an experiment could be adjusted by finding the optimal balance between hydrogen bond network and ionic structure for a certain application. Of course, the slit sizes corresponding to minima in hydrogen bonds or maxima in the density will change with the IL. For our simulations of EAN, we find that every 0.4 nm, starting from a slit size of 1.1 nm, a new layer appears in the system. It is obvious that this interval will be proportional to the molecular size of the IL components, i.e. ILs with larger anions and cations will induce more

spread ionic peaks in the density profiles, and so more distance between ionic layers. However, more studies with different ILs need to be done to find the exact dependence.

As it was mentioned, it is well established that the molecular structure in confined systems is closely related with the ionic dynamical properties, e.g. ionic conductivity or ionic diffusion. We try to shed some light in this field calculating the ionic self-diffusion coefficients for these simulations from the MSDs. In Fig. 9 we represent these MSDs, as well as the fitting parameters of Eq. 2 (D and σ). A first appreciation that could be done looking at the MSDs is that, as expected, they recover the MSD of ions in the bulk as the slit size increases. Note that now the MSD for the system with 8 nm of wall separation and a lateral box size around 9.5 nm takes lower values compared with the bulk one throughout the whole simulated interval. This was not the case for the same system with only 1000 ionic pairs presented in Fig. 4. This constitutes a new evidence that the dynamics of these mixtures are significantly affected by the size of the simulation box. In these simulations we observe the same general trend previously reported: the ionic mobility increases with the distance between walls. This trend is also reflected in the diffusion coefficients. However, there is no clear correlation between these coefficients and structural properties. It is true that there are two local maxima of the diffusion coefficients at slit sizes of 1.5 nm and 1.9 nm, which correspond to maxima in the densities and minima in the number of hydrogen bonds. Besides, there is another local maximum for slit sizes of 2.5 nm that does not follow the same rule. Our main conclusion derived from these diffusion coefficients is that some degree of correlation between the long-time dynamics and the confinement conditions seems to exist, but it must be carefully analyzed. To obtain more reliable results longer simulation times are needed. On the other hand, although diffusion coefficients do not show any clear correlation with the structural magnitudes, the fluctuations in the positions of the anions at short times do. We see how the systems with higher densities (and so with the best defined ionic layering), and thus with average number of hydrogen bonds close to their bulk values, present very small position fluctuations, i.e. the ions are more strongly bounded to a reference position. On the other hand, the maxima in the values of σ , which are very close to its value in bulk systems ($\sigma_{bulk} \sim 1.4 \text{ \AA}$), match with maxima in the number of hydrogen bonds, where the electrostatic network is relaxed and more mobility is allowed. Definitely, what we can assert is that the short-time dynamics is directly affected by the structural properties.

Figure 9: (a) Anions' MSDs for the simulations with constant lateral box size. (b) and (c) summarize the dependence of the diffusion coefficients and position fluctuations obtained in the same way as for Fig. 4. The results for bulk simulation are also represented in black dashed lines for comparison.

To provide a new evidence that corroborates the relation between short-time dynamics and structure we calculated the VACFs for the anions as

$$\text{VACF} = \frac{\langle \vec{v}(t) \cdot \vec{v}(0) \rangle}{\langle \vec{v}(0) \cdot \vec{v}(0) \rangle}, \quad (3)$$

where $\vec{v}(t)$ is the velocity of the center of mass of the molecule at time t . For this calculation we used the trajectory from the simulations of 50 ps recording the velocities each fs. These VACFs are represented in Fig. 10. To see the differences between the curves depending on the slit size we also represent the average collision time, i.e. the shortest time at which the VACFs start taking negative values due to an inversion in the molecular motion. Observing these collision times we see that they follow exactly the same trend as the hydrogen bonds, which at the same time are directly related to σ . This makes sense because, as it was mentioned, the systems with the lowest number of hydrogen bonds are those with the highest densities, so the average distance between molecules is shorter and then the average time for a molecule to collide with one of its nearest neighbor is consequently shorter.

Figure 10: Anions' VACFs for the simulations with constant lateral box size. The inset represents the dependence of the collision time with the slit size of the simulation, i.e. the intersection of VACFs with the small black line placed at around $t = 0.12$ ps.

Although a clear correlation between the slit size and long-time dynamics can not be studied when the most confined systems are analyzed, we saw that these coefficients increase with the distance between walls, going from very low values for systems with slit size around 1 nm to values very close to the ones for bulk systems in the less confined geometries with slit sizes of 6 nm or 8 nm. This suggests the possibility of two ionic populations in our simulations, one close to the interfaces with very low mobility and the other in the center of the simulation box (bulk region) with the same mobility as in bulk simulations. This behavior has been already observed for ILs

inside carbon nanotubes [38] and inside silica and carbon mesopores [68]. In these systems, the innermost layer of the ionic double layer close to the interface shows lower mobilities than the second one and this in turn is lower than the third one. To see if this is the case for planar interfaces, we calculated the MSDs for the anions in the first three layers of the system with a slit size of 8 nm. To do that, we needed to know which anions belonged to each layer, so we performed an initial classification in the first simulation frame based on the positions of their center of mass in the box. Then, we iterated over the whole trajectory and removed the anions which moved between layers from this classification, so at the end we obtained a list of the anions that stayed in the same layer during the whole simulation. The number of anions that stayed in each layer during the whole simulation is summarized in Table 1. The obtained MSDs are represented in Fig. 11 together with a plot which summarizes the limits of each region. It has to be mentioned that the represented MSDs in case of layers close to the walls are the result of the average between the anions MSDs close to both left and right walls. Firstly, we can see that the anions in the central region have a larger mobility than those close to the interfaces. Secondly, it can be observed that the anions in the first two layers close to the walls are basically stacked (in a solid-like phase) while those in the third one are more mobile (in a more fluid-like phase). Finally, the behavior of anions in the central region is very similar to that in bulk simulations. In summary, the mobility of ions close to planar interfaces follows the same pattern as in other geometries, e.g. nano-tubes [38] or mesopores [68].

Table 1: Number of anions in each layer of the simulation with a slit size of 8 nm during the whole simulation.

Layer	No. Anions
1 st left	60
1 st right	42
2 nd left	125
2 nd right	90
3 th left	266
3 th right	294
Bulk	2065

Figure 11: Top: the MSDs for the anions in different layers in the simulation with slit size of 8 nm.

Bottom: the anion's density profile with shadowed regions delimiting the anions layers.

We may use our simulations to compute the range of the wall effects in the ionic mobility to see how many ionic layers are affected by the presence of the wall. To do that, we assume that a system with the walls separated by a distance L can be described as the sum of two subsystems, a low ionic mobility one close to the walls, and another one in the middle of the box with the same dynamical behavior as in the bulk. Of course, L should be large enough to allow the system to contain both regions. If we denote by z_C the width of each simulation slice representing the region with low ionic mobility, then the width of the bulk region can be calculated as: $z_B = L - 2z_C$. Now, if we remove this bulk region and stack together both low mobility slices, the system with walls separated by L can be thought of as the sum of a bulk and a confined system with a wall separation of $2z_C$ with low ionic mobility (see Fig. S1 in the SI for a schematic representation of this description for the simulation with 8 nm of wall separations). In this scenario we can calculate the anions MSD of the system as the weighted average of the MSDs of each subsystem:

$$\text{MSD}_L = \frac{L - 2z_C}{L} \text{MSD}_B + \frac{2z_C}{L} \text{MSD}_C. \quad (4)$$

We can evaluate this MSD_L for multiple z_C using the MSD_C of the corresponding simulation with wall separations of $2z_C$ and compare it with the MSD_L calculated directly from the simulation. This comparison for the case of $L = 8$ nm is represented for all the simulations with slit sizes between 1 nm and 3 nm in Fig. 12. To have a better insight of how much the curves differ from each other we calculated the average normalized distance between the MSD obtained from Eq. 4 and that found directly from the simulation. We calculated this average between $t = 10$ ns and $t = 20$ ns and they are represented also in Fig. 12. We can see that if we consider the region affected by the wall to have a width of $z_C = 0.5$ nm, i.e. we use the MSD_C calculated from the system with slit size of 1 nm, the MSD_{8nm} obtained with Eq. 4 is quite far from that obtained from the simulation (average normalized distance around 0.08). However, if we consider $z_C \geq 2.7$ nm both the test and real MSDs are almost identical (normalized distance ca. 0.005). In the light of these results we can say that in a first approximation and for our studied IL (EAN) the region affected by the presence of walls extends around 1.35 nm from the walls. Looking at the ionic

density profiles this distance corresponds with the end of the third layer of anions. This simple mean-field-like theory can be easily extended to describe a wide variety of systems, from different ILs to more exotic confinement conditions (spheres, nano-tubes...). Moreover, the estimation of this characteristic length associated to the range of the effects of the interfaces in the IL can be very interesting from the experimental point of view as it provides an idea of the length scale at which the IL properties are expected to start changing due to the confinement. This is helpful to choose the appropriate distribution of ion pore sizes in situations where it is essential to control it (ionogel matrices, nano-porous supercapacitors or electrodes...).

Figure 12: Comparison between the MSD for the system with wall separation of 8 nm and those obtained from confined simulations and Eq. 4. Left: the whole curves for the estimated MSDs, the one calculated from the simulation (solid black) and the MSD for anions in bulk (dashed black) used in Eq. 4. Right: the average normalized distance between each test MSD and that calculated from the simulation between $t = 10$ ns and $t = 20$ ns.

4. Summary and Conclusion

We performed MD simulations of EAN IL confined between two implicit walls with the parameters of carbon in the NpT ensemble to study the properties of this liquid under nano-confinement conditions. In a first set of simulations we study systems with distances between walls from 1 nm to 6 nm keeping the number of ionic pairs constant (set to 500) and also a system with 8 nm of wall separation and 1000 ionic pairs (to avoid very short lateral box sides). In these simulations we observed that structural properties such as the global number density or the potential energy are directly related with the ionic layer formation between the interfaces. The general trend is that the density and the potential energy of the systems take lower values as the distance between walls decreases. Moreover, studying these quantities in simulations where the distance between walls is slightly increased (in steps of 1 Å), it is observed that the mentioned global trend hides an underlying phenomenon related with the axial ionic structure. Indeed, when the distance between walls allows well defined ionic layering, i.e. when there are very well defined peaks in the axial ionic density profiles, the global system density takes maximum values, which is the result of a highly packed ionic configuration. We also tried to find a correlation between these peaks in the structural properties and the dynamical behavior of the components. To do that, we

calculated the MSDs of anions and linearly fit them to obtain the self-diffusion coefficients as well as the spatial fluctuations in the position of anions at short times. We were also able to see a general trend in the diffusion coefficients, i.e. the most confined simulations showed the lowest ionic mobilities while as the distance between walls increases the bulk properties are recovered. However, the fluctuations of these dynamical quantities at low slit sizes did not show a clear dependence with the ionic structure.

To check if the lateral simulation box size influenced the system properties, we performed a new set of simulations with constant slit size (1.7 nm) changing the number of ionic pairs to obtain different constant for all the simulations but the dynamical ones do not. Moreover, we found that the dependence of the obtained diffusion coefficients can not be modeled by the hydrodynamic interaction between the simulation components and their replicas. At the light of those simulations we conclude that a good choice for the lateral simulation box size to obtain reliable dynamical properties and keep a reasonable computational cost could be around 9.5 nm. However, although this was a good choice for our simulations, it should be fitted depending on the chosen IL or the distance between walls.

With this lateral box size in mind we performed a final set of simulations with slit sizes from 1 nm to 3 nm in steps of 0.1 nm but fitting the number of ionic pairs to obtain that size. We repeat the structural analysis for these simulations and we recovered the results from the first set. Moreover, we obtain more detailed description of the ionic layering in this slit size range. We extended the structural study and calculated the average number of hydrogen bonds between the simulation components. This quantity also shows a dependence with the ionic layering. In general, the ions form more hydrogen bonds (comparing with bulk) when they are confined. However, minima of this magnitude (values close to that in bulk simulations) are registered when the global system density is a maximum, i.e. when the IL is in a crystal-like phase. Regarding the dynamical properties, with these new set of simulations we observed that some local maxima in the self-diffusion coefficients are registered for the systems with the highest densities. Nevertheless, for a better insight in this phenomenon, simulations with larger times and more suitable force-fields are needed to obtain more reliable results for the ionic diffusion.

Finally, we see how the MSDs of the anions are changed as they are closer to interfaces. For this task, we calculated the MSDs for anions in different pairs to obtain the same lateral box size as the previous simulations. We reported that the anions in the first two ionic layers are less

mobile than those in the third one and this in turn is less mobile than those in the bulk region. This behavior was already reported for ILs under confinement in other kind of geometries [38, 68]. Moreover, with a simple theory we obtained that for EAN the affected region by the presence of interfaces extends around 1.35 nm from each wall which, in our simulations, matches with the end of the third anion's layer. This theory can be applied in other ILs and geometries to estimate the characteristic length at which the confinement conditions start to affect the IL properties, which can be very useful from both the theoretical and experimental points of view.

5. Conflict of Interest

There are no conflicts to declare.

6. Acknowledgments

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Credit author Statement

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Oscar Cabeza: Writing (all), Supervision, Funding acquisition

Luis J. Gallego: Writing (all), Supervision.

Luis M. Varela: Conceptualization, Investigation, Methodology, Writing (all), Supervision, Project administration, Funding acquisition

Declaration of interests

- ☒ The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.
- ☐ The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

HIGHLIGHTS

- Ethylammonium nitrate ionic liquid under ultra-nanoconfinement between planar interfaces.
- Strong correlation between structure and dynamics.
- Structure determined by ion packing, hydrogen bonding, electrostatic and van der Waals forces.
- Slower dynamics for very low slit sizes (below 3 nm).
- The spatial range of the interfacial effects in the ionic liquid dynamics is estimated to be 1.35 nm for EAN.

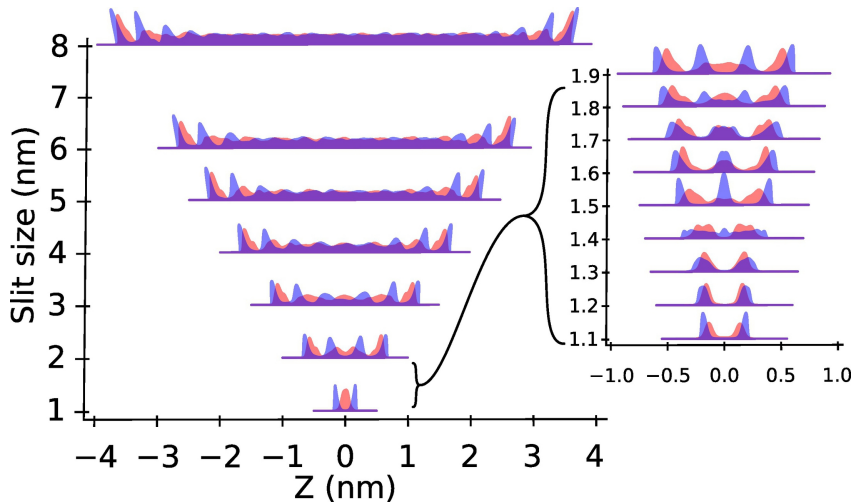


Figure 1

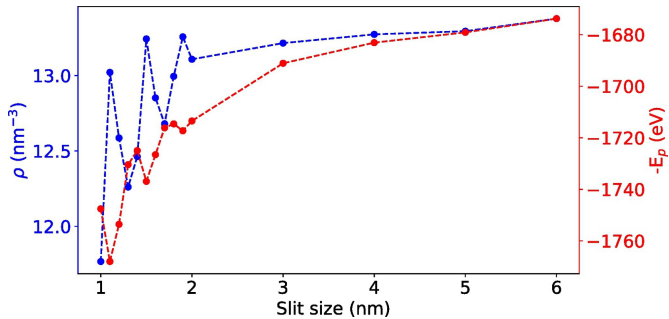


Figure 2

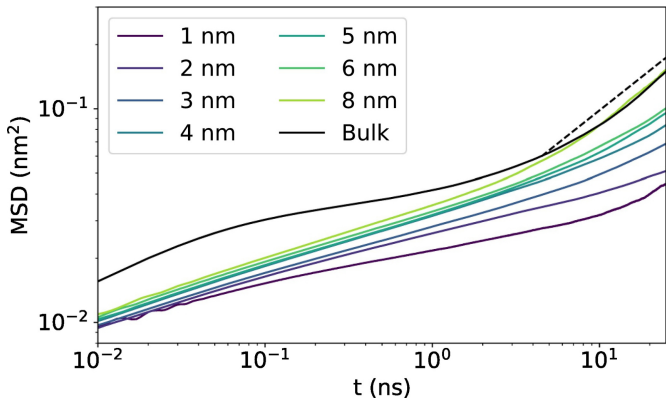


Figure 3

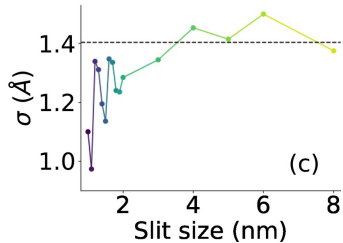
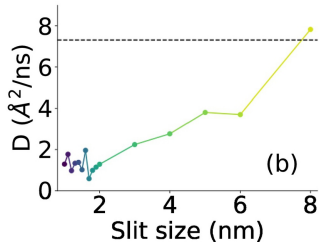
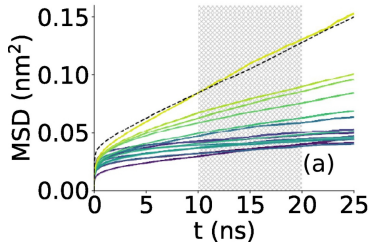
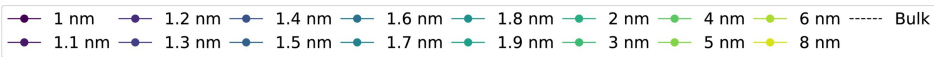


Figure 4

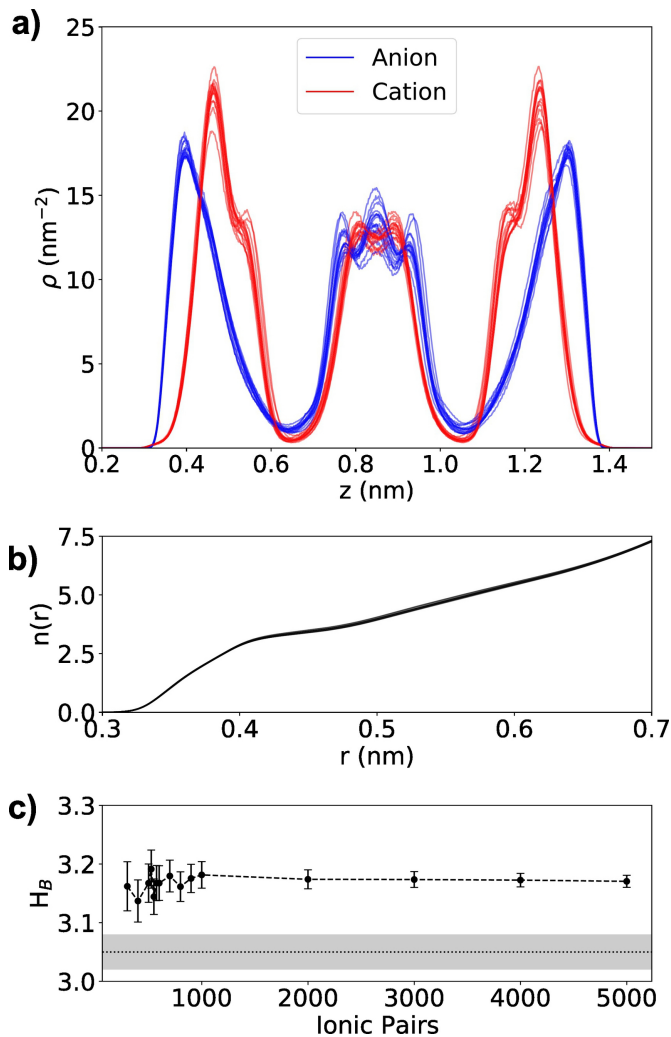


Figure 5

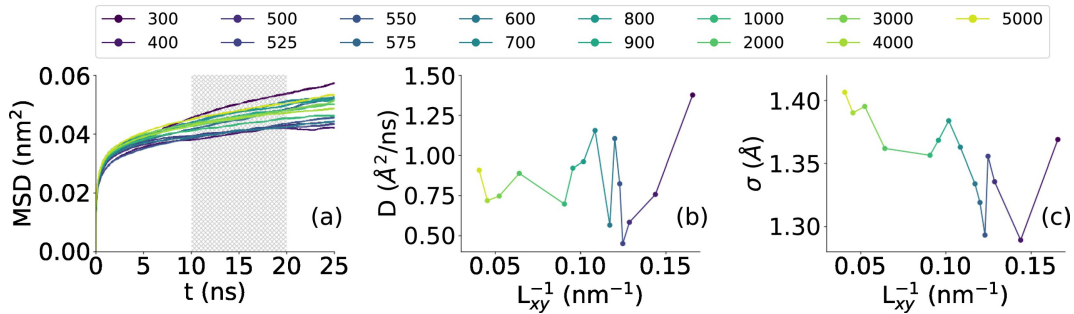


Figure 6

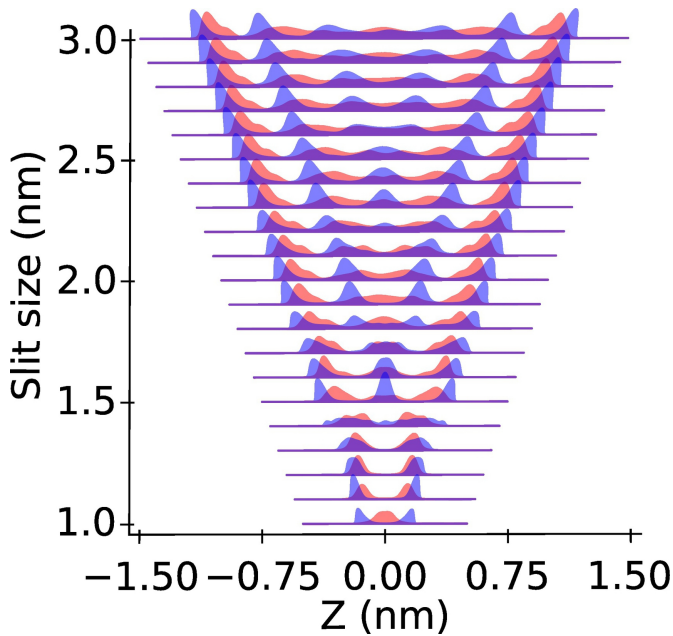


Figure 7

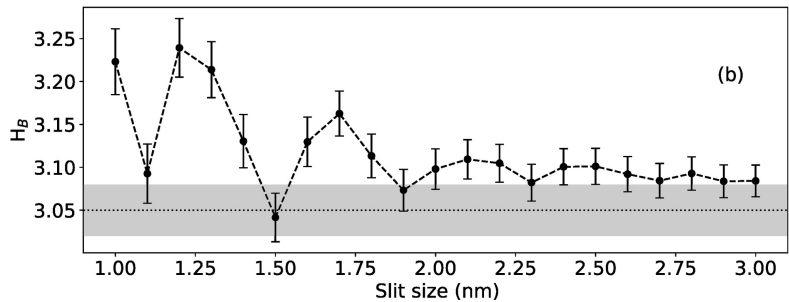
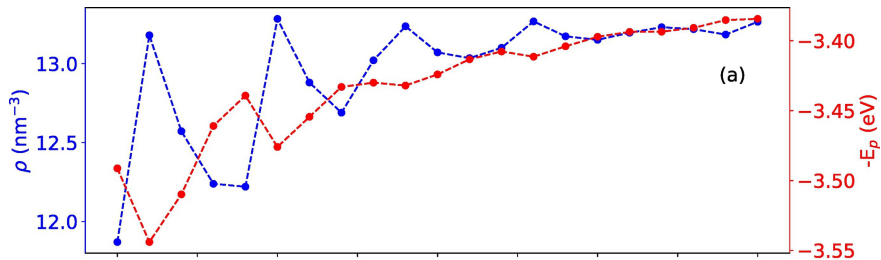


Figure 8

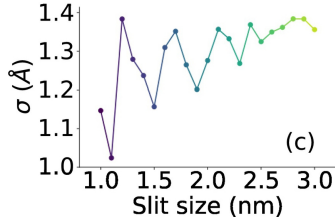
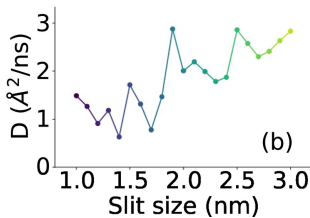
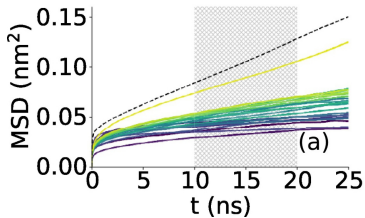
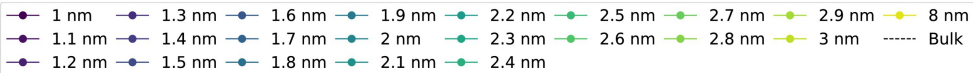


Figure 9

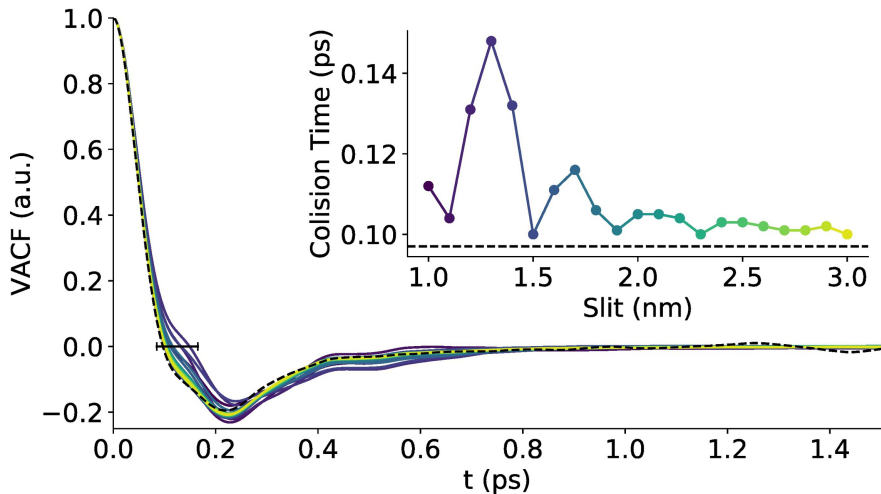
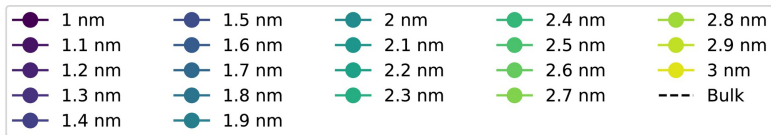


Figure 10

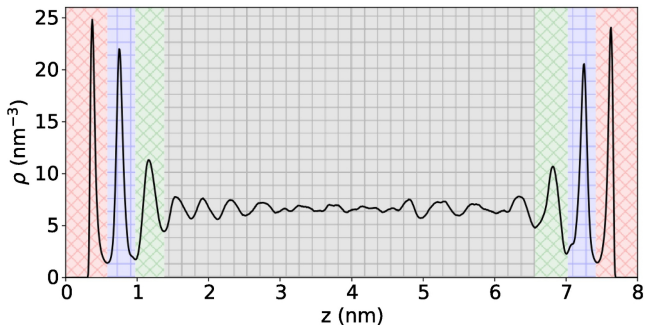
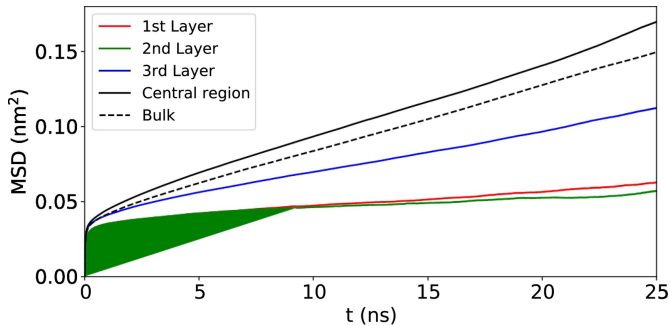


Figure 11

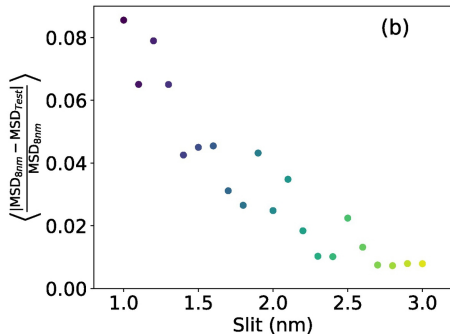
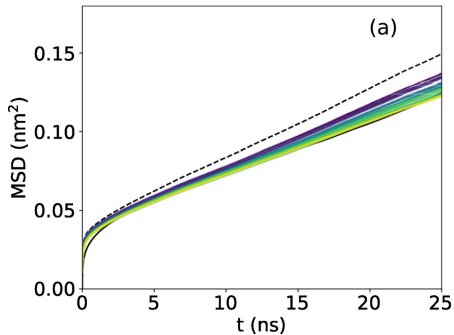


Figure 12